USCD Machine Learning Bootcamp Capstone Architecture Writeup

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Using a Large Language Model (LLM), I created an Angular app that tutors its users on the Swedish language for my capstone project. The app's source code consists of three components: one to select the topic, one to answer a question and receive feedback, and one to score the quality of the feedback provided. Through a series of prompts to one of four LLMs, the app questions the user in Swedish, awaits a Swedish response from the user, and then delivers feedback in English on each of the user's mistakes. The app supports Mistral Large 3 [1], Ministral 3B [2], Gemma 3 27B PT [3], and Gemma 3 4B IT [4]; the user may pick one of these via a select dropdown at the top of the app.

AWS Amplify simplifies the deployment process. It requires only my GitHub repo and my consent in order to produce a publicly accessible link. Upon deployment, it provides access logs and supports integration with AWS CloudWatch for monitoring. AWS Amplify charges me a nominal fee for the duration in which the app stays up, and I can take it down at any time.

All users begin in the topic selection component. It has a predefined list of topics (eg. time, nature, etc.) from which a user may select. It also contains a form field for any user with a specific Swedish question in mind. The program sends the topic through an EventEmitter to the next component. If the user selects a predefined topic, then the program sends it to the user's chosen LLM in the following prompt: "Please ask me a question in Swedish about [topic]." The program provides any specific Swedish question from the user verbatim without a prompt.

The next component begins with a Swedish question (prompted or specific) posed to the user. There is a form field for the user to answer the question in Swedish and a button to request feedback. This component stores both the question and the response internally. When the user requests feedback, the program sends the user's chosen LLM the following prompt: "Given the question of [question], please provide feedback in English to the spelling and grammatical mistakes of each word in the following user response: [response]." Once the component displays the feedback on the screen, the user may either request a follow-up question or go back to the previous component and change the topic.

For a full example of this process, I submitted the predefined topic of "time" and received the following question: "Hur skulle du beskriva din relation till tid? Känner du att du har gott om tid, eller önskar du att dygnet hade fler timmar? Varför?" This translates to, "How would you describe your relationship to time - Do you feel that you have enough time or do you wish the day had more hours? Why?" I provided the following response: "Jag önskar ofta att klokan har fler timmar eftersom jag har för många hobbyer." (I misspelled "klokan" on purpose.) I received the following feedback:

1. klokan -> klockan (or "dygnet") - Klokan means "the wise one" (eg. a wise person or owl), but you meant "klockan" (the clock) or "dygnet" (the day/24-hour period). Correct: "Jag

önskar ofta att klockan hade fler timmar…” (Or more naturally: “Jag önskar ofta att dygnet hade fler timmar…”)

2. har -> hade (subjunctive mood) - Since this is a hypothetical wish, Swedish uses the **subjunctive** (“hade instead of “har”). Correct: “...att klockan **hade** fler timmar…”

3. Grammar note (optional improvement): “The phrase “för många hobbyer” is grammatically correct, but you could also say: “Jag har **alldeles för många hobbyer”* (emphasizes “way too many”). “Jag har **så många** hobbyer att tiden inte räcker till* (“I have so many hobbies that time isn’t enough”). Fully corrected version: “Jag önskar ofta att dygnet hade fler timmar eftersom jag har för många hobbyer.” (Or, more naturally, “Jag önskar ofta att dygnet hade fler timmar - jag har alldeles för många hobbyer!”)

4. Additional notes: “Fler” vs. “mer”: “Fler” is used for countable things (e.g., “fler timmar” = “more hours”). “Mer” is used for uncountable things (e.g., “mer tid” = “more time”). Your original use of “fler” is correct here! Why “hade”? Swedish uses the past tense (“hade”) for hypotheticals/wishes, similar to English: “I wish I had more time” -> “Jag önskar att jag **hade** mer tid.” Your response was otherwise clear and natural! Let me know if you’d like help refining it further.

This feedback is the result of two prompts, each sent by one of the app’s components. The LLM returns either a response saved by the app as an internal variable or an error message printed to the Browser Inspect Window for debugging purposes.

Upon receiving feedback in the second component of the app, a user may proceed to the feedback scoring page, which prompts for a feedback “answer key”. The app will then grade the feedback from the user’s chosen LLM against the user-provided “answer key” on a score from 1 to 10. The user may also skip to the scoring page immediately to score any specific feedback. To obtain the score, the app sends the LLM the following prompt: “Given the AI-provided feedback of [feedback] and the feedback answer key of [correct-feedback], please provide a score of 1 to 10 to measure the semantic accuracy of the AI-provided feedback. Please begin the response immediately with the score before going into any reasons or justification.” The scoring page also accepts a JSON file through which the user’s chosen LLM can provide feedback to and score a list of questions and answers. It is through this page that I assessed the quality of each of the four LLMs using testing data of my own creation.

When building a machine learning model, a developer typically requires both training data and testing data. Training data teaches the model the pattern recognition it needs to assess a specific problem while testing data measures its effectiveness. An LLM does not require training; however, as it is already pretrained. Furthermore, no suitable testing data is available online. Although Språkbanken contains dozens of question and answer datasets in Swedish (including “Super-LIM” [5]), none include feedback statements addressed towards deliberately incorrectly written Swedish sentences (eg. users who forgot to add “på” in “Jag tittar på TV:n.”). Therefore, I was required to create my own testing data. In the repo, the source code includes a json file of 100 question / answer / feedback triplets obtained with assistance from ChatGPT.

Lastly, when the app scores a list of question / answer / feedback triplets provided from a JSON file, it provides the mean, median, and standard deviation for the list of resulting scores.

Works Referenced

[1] Mistral Large 3. https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-mistral-ai-mistral-large-3.html

[2] Ministral 3B. https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-mistral-ai-ministral-3b.html

[3] Gemma 3 27B PT. https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-google-gemma-3-27b-pt.html

[4] Gemma 3 4B IT. https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-google-gemma-3-4b-it.html

[5] “Super-LIM”. https://spraakbanken.gu.se/resurser/superlim